TBXT Hit Identification

End-to-end virtual screening pipeline

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Target: human TBXT (Brachyury) DBD, residues 42–219

Source: onepot 3.4B CORE catalog

Submission: 4 ranked SMILES across pockets A, F, G

Strategy

"Traditional" modern virtual screening approach

Every signal on this target is weak. Stack them; filter aggressively.

Weak, independent signals:

- Zenodo SPR — ~0.65 AUROC ceiling
- Boltz — pose OK at 3/4 sites
- Vina — classical, false positives
- Fragment similarity — uneven
- Physchem + risk — necessary only

Stack them:

1. Zenodo → ranker, not predictor
2. Boltz → pose, not raw ΔG
3. Vina → orthogonal + anchor check
4. Physchem + PAINS + synthesis risk throughout

Target and pocket plan

TBXT / Brachyury — T-box TF, key dependency in chordoma
DNA-binding domain (res 42–219), PDB 6F59
SGC TEP identifies 5 fragment pockets; we target **A, F, G**

TEP pocket	Anchor residues	Frags	Slots
A / A'	R180, V123, L91, I125, V173, I182	26	2
G	R54, E48, E50, L51, K76	10	1
F	Y88, D177, V173, I182	4	1 (spec)
~~**D**~~	G112, H100, P115	5	0 (bad Boltz pose)

Why these pockets (superposed all PDBs, scored contacts vs Newman signatures):

- **A (2 slots)** — most de-risked: 26 fragment hits, the only site Newman progressed to µM SPR (thiazole → 8A7N, 14–20 µM). Clean Boltz pose vs 5QS2.
- **G (1 slot)** — 10 hits, largest G177D hotspot. Polar pocket near C-term helix; DNA-interface-adjacent angle distinct from A.
- **F (1 slot, speculative)** — novel MoA: Y88 is the P300-interaction residue; pocket mediates KDM6 co-activator recruitment. Engages G177D variant directly. Induced, buried → oracle weaker, so lean on fragment similarity.
- **D (dropped)** — Boltz pose tilted vs 5QS0 crystal, only 4–5 hits, DNA-competitive angle already covered by G.

Pocket IDs follow the SGC TEP datasheet.

Modeling the Zenodo SPR data

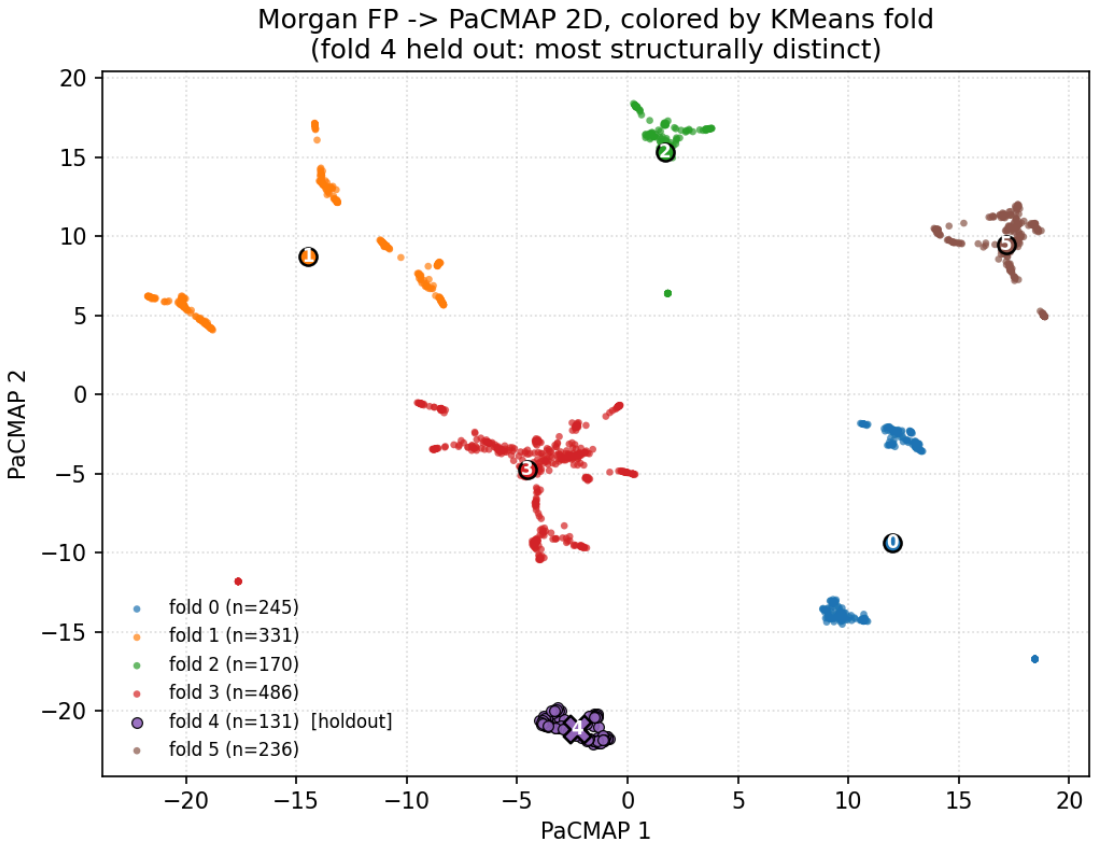
Data: 2,143 SPR records, 14 batches (Oct 2020 – Jan 2023), 1,545 unique compounds after cleaning.

The dataset was hard to model.

- Batch date alone explains $R^2 \approx 0.15$ of pKD variance
- ~30% activity cliffs at Tanimoto ≥ 0.7; replicate std ≈ 0.92 pKD (~100× KD)
- Regression: all attempts negative R^2 — data ceiling, not model ceiling
- Tried **CheMeleon chemistry foundation model**; competitive signal but kicked out under time pressure

What worked: reframe as a binary ranker.

- **Excluded mid-range KD** (pKD 3–5 gray zone) — unlearnable with ~0.9 pKD replicate noise
- **PaCMAP + KMeans6 folds** — structurally-distinct splits, not scaffold-leaky Butina
- **12-booster XGBoost ensemble** (2 fingerprints × 6 leave-one-fold-out) on the cleaned label



Training compounds in PaCMAP space, colored by KMeans6 fold assignment. Structurally-distinct clusters — the ensemble trains 12 boosters leave-one-fold-out so every compound is scored by a booster that never saw it.

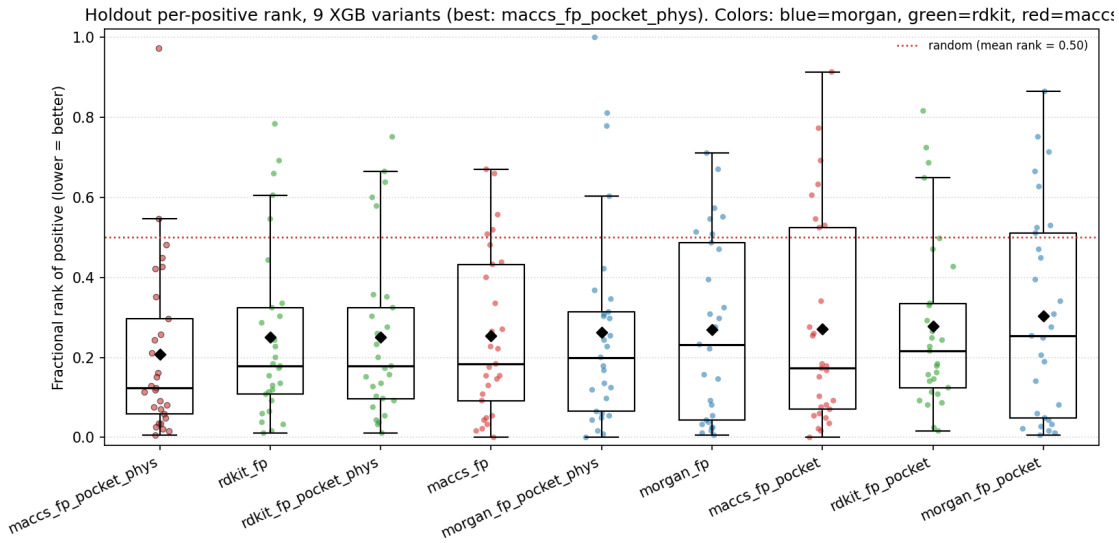
Model performance — positive ranking

The right metric for a ranker is: where do the true positives land?

Evaluated on the try4 holdout (fold 3, n=186, 29 positives, 15.6% prevalence). For each positive, compute its **fractional rank** among the 186 holdout compounds after sorting by predicted score. Random = 0.5, perfect = 0.

- `maccs_fp_pocket_phys` (the deployment recipe): **median rank 0.124, mean 0.208** — true positives land in the top ~12% on average
- Tightest distribution of all 9 variants tested
- Only variant where non-FP features materially contribute (86% MACCS / 6% pocket / 8% physchem importance)

Morgan sharpens the *very top* of the ranking but has a long right tail. MACCS+pocket+physchem is wider at the top and has no catastrophic misses. Stacking both captures both behaviors.

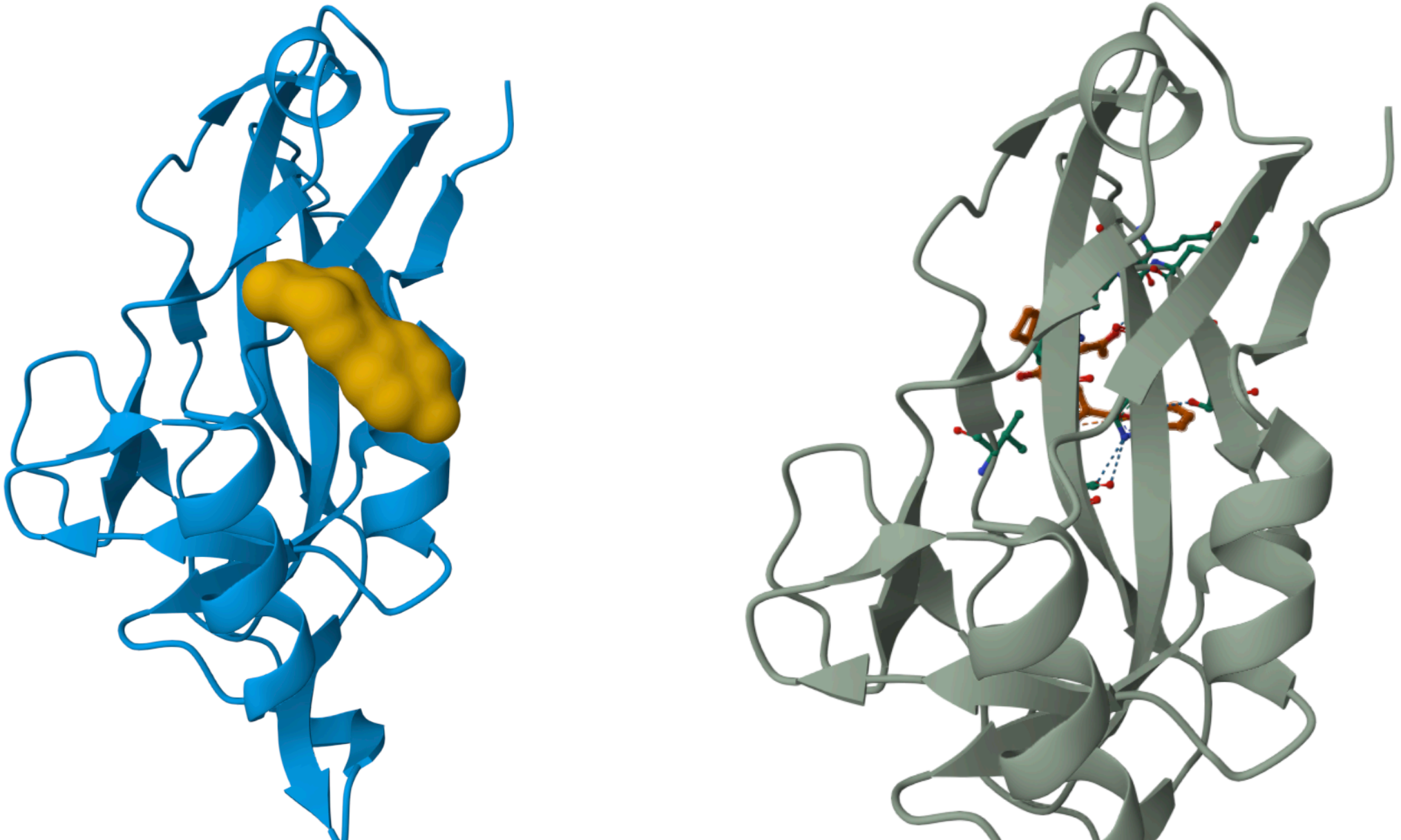


Per-positive fractional rank across 9 fingerprint × feature-set variants, holdout fold 3 (29 positives). Lower = better; random = 0.5 (grey line).

Pipeline — end-to-end with numbers

```
onepot CORE (3.4B)                  SGC TEP fragments (45)
       │                                      │
       ▼                                      │
[1] Downfilter: regex → RDKit                 │
    substructure + physchem + PAINS + risk    │
       │                                      │
       ▼                                      ▼
  897,431 survivors     [1b] onepot API neighbor queries
       │                     + physchem + PAINS + risk
       │                     51,560 unique compounds
       └──────────────────────────────────────┘
                         ▼
        Combined screening library: 178,332
                         ▼
      [2] XGBoost 12-booster ensemble → p_binder rank
                         ▼
      [3] Pocket-assign → split A / F / G (top 5000 each)
                         ▼
      [4] Diversity cluster (Morgan, Butina-style)
                         ▼
      [5] Boltz pose prediction + pose QC
            (site-localization rejects ~90–95%)
            survivors: <100 per pocket
                         ▼
      [5b] Exclude Zenodo-similar compounds
            (ECFP4 Tanimoto > 0.85 to any SPR compound)
                         ▼
      [6] AutoDock Vina + anchor-residue check
            R180 (A) · E48/R54 (G) · Y88 (F)
                         ▼
              Final 4 (2·A, 1·G, 1·F)
```

Boltz site-localization filter — in action



Recommended compounds *(mock — to be populated)*

Rank	Pocket	SMILES	Scaffold	Rationale
1	A	`TBD`	thiazole-acetamide	Top XGBoost rank in A; Boltz pose recapitulates 8A7N R180 H-bond; Vina –8.9 kcal/mol.
2	A	`TBD`	morpholino-thiazole	Scaffold-diverse A-pocket hit; strong R180 + L91/V123 hydrophobic cluster engagement.
3	G	`TBD`	aryl sulfonamide	Polar head engages E48 + R54; highest-ranked G-pocket survivor after pose QC.
4	F	`TBD`	compact bicyclic	Speculative novel-MoA slot; Y88 contact + D177 proximity; fragment-Tanimoto-led.

Each row is backed by convergent evidence: XGBoost `p_binder`, fragment Tanimoto ≥ 0.35 in assigned pocket, Boltz pose matching Newman co-crystal, Vina score within 1 kcal/mol of per-pocket best, anchor-residue contact, Chordoma physchem satisfied, Max Tanimoto to Naar set < 0.6.

Appendices

Appendix A — Full submission evidence table *(mock)*

Rank	onepot ID	SMILES	Pocket	Scaffold	Boltz ΔG (kcal/mol)	Vina (kcal/mol)	`p_binder`	Anchor contact	Rationale
1	`OP-000000001`	`TBD`	A	thiazole-acetamide	−8.4	−8.9	0.87	R180 H-bond (2.8 Å)	Top XGBoost rank A; reproduces 8A7N pose; L91/V123/I125/V173 hydrophobic contact cluster.
2	`OP-000000002`	`TBD`	A	morpholino-thiazole	−8.1	−8.5	0.81	R180 H-bond (3.0 Å)	Scaffold-distinct A-pocket hit; morpholine O accepts from R180 guanidinium; clean Boltz pose vs 5QSD.
3	`OP-000000003`	`TBD`	G	aryl sulfonamide	−7.6	−7.9	0.74	E48 (salt bridge) · R54 (3.1 Å)	Polar head reaches E48/R54 hotspot; pose QC passes vs 5QS6; DNA-interface-adjacent vector.
4	`OP-000000004`	`TBD`	F	compact bicyclic	−8.8*	−7.3	0.63	Y88 (3.2 Å) · D177 proximity	Novel-MoA speculative slot; high fragment Tanimoto to K2P (5QSA); MW < 400 fits buried cavity.

*Pocket F Boltz ΔG is inflated by burial + induced-fit and is used as pose-only signal, not absolute score.

All 4 satisfy: Chordoma physchem (LogP ≤ 6, HBD ≤ 6, HBA ≤ 12, MW ≤ 600), max Tanimoto to Naar set < 0.6, and 4 distinct Bemis–Murcko scaffolds.

Appendix B — Stage 1 + 1b details (library build)

Downfilter 3.4B catalog

`scripts/downfilter_onepot/`

1. Coarse SMILES regex against canonical + Kekulized fragment SMILES + Bemis–Murcko scaffolds
2. RDKit worker pool: PAINS + bad-SMARTS + `HasSubstructMatch` vs 45 fragments
3. Hard filters: MW/LogP/HBD/HBA, 10–30 HAC, ≤ 5 rings, ≤ 2 fused, ≥ 1 fragment hit
4. Sharded parquet, rollup every 5M rows

Output: 897,431 compounds

onepot API nearest-neighbor queries

`scripts/query_onepot_neighbors.py`

- Per query pocket (A / F / G): 1 API call per SGC fragment
- Pocket A: 1000 neighbors/frag
- Pocket D/F/G: 2000 neighbors/frag
- ECFP4 (r=2, 2048) → route to best pocket
- Physchem + PAINS + synthesis-risk filters applied
- Fragment-substructure filter skipped (already similarity-selected)

Output: 51,560 unique compounds

Combined: `data/screening_library_combined.csv` **— 178,332 compounds**

Appendix C — Model metrics deep-dive

Deployment ensemble — 12 XGB boosters (2 FP × 6 leave-one-fold-out splits on PaCMAP-KMeans6).

Overall OOF (N=708, 12.3% prev):

- AUROC **0.764** · AUPRC **0.337** (random 0.123)
- Top-30 precision on try4 holdout: **17/30** = 57% (random ~16%)
- Mean fractional rank of positives: **0.208** (median 0.124)

Label choice: pKD > 5, filter 3–5 gray zone.
Try1 (no filter, top-quartile): holdout AUROC 0.48 (random).
Try2 (filter gray zone): +0.06 to +0.12 OOF AUROC across variants.
Try4 (ablated FP × features): **maccs+pocket+physchem won** — only variant where non-FP features materially contribute (86% MACCS, 6% pocket, 8% physchem importance).

Why PaCMAP folds: Butina folds gave a deceptive 97% AUROC / 89% precision — scaffold leakage. PaCMAP breaks scaffold-level correlation and gives honest generalization estimates.

Feature pipeline (183-dim):

Slice	Source
167 cols	MACCS keys
8 cols	Pocket scores (Tanimoto + substruct, 4 pockets)
8 cols	Physchem (MW, LogP, HBD, HBA, HAC, rings, TPSA, rotB)

Ensemble probabilities used as **rank score only**, not calibrated P(binder).

Appendix C2 — Per-fold metrics

Morgan XGB booster family (per-fold OOF):

Held-out fold	N train	N test	Pos train	Pos test	OOF AUROC	OOF AUPRC
0	596	112	81	6	0.461	0.101
1	563	145	48	39	0.648	0.389
2	630	78	81	6	0.551	0.100
3	522	186	58	29	**0.800**	**0.410**
4	646	62	86	1	0.918*	0.167
5	583	125	81	6	0.637	0.282
Overall	—	708	—	87	**0.736**	**0.312**

*Fold 4 AUROC computed on 1 positive — uninterpretable in isolation. Fold 3 (29 positives) is the best-powered single-fold estimate.

Per-fold variance is extreme because positives are unevenly distributed across folds. Fold 0 (6 positives, AUROC 0.46) is a structurally distinct chemistry cluster the others don't sample well; the ensemble mean compensates. This is the honest cost of structure-aware splits on a small, batch-confounded dataset.

Appendix C3 — Pocket assignment details

Appendix D — Stage 4 + 5 (cluster + Boltz)

Top 5000 per pocket → diversity cluster

- Rank by `p_binder_ensemble_mean`
- Butina-style clustering on Morgan FP
- Kills redundant scaffolds *before* the expensive oracle runs

Model score is cheap on 178k. Boltz is the bottleneck — wasting it on scaffold twins shrinks effective pocket coverage.

Boltz pose prediction (Boltz Lab cloud)

Ran Boltz on the clustered top-5000-per-pocket. Pose QC is strict — the site-localization filter alone rejects **~90–95%** of Boltz outputs.

Post-QC survivors docked with Vina:

- **Pocket A: <100**
- **Pocket G: <100**
- **Pocket F: <100**

Pocket D not run (pose tilted vs crystal).

Site-localization filter: reject any compound whose Boltz pose places the ligand outside the intended pocket, even when induced-fit displacement is allowed. This is the dominant filter; catches off-site binders the affinity score alone would not flag.

Per-site cautions carried into ranking:

Appendix E — Stage 6 + 7 (Vina + final selection)

AutoDock Vina · `docs/docking.md`

- Per-pocket receptor PDBs, box centered on reference fragment, 10 Å padding
- Multi-conformer, interaction analysis (H-bonds, π-stacking, hydrophobic, salt bridges)

Anchor-residue hard filter:

Pocket	Required contact
A	H-bond to **R180**
G	Contact to **E48 or R54**
F	Contact to **Y88** (±D177)

Fail the anchor check → drop regardless of Vina score.

Final 4 — convergent evidence required:

- ✅ XGBoost `p_binder_mean` ranks high
- ✅ Fragment Tanimoto ≥ 0.35 in assigned pocket
- ✅ Boltz pose resembles crystal fragment
- ✅ Vina score within 1 kcal/mol of per-pocket best
- ✅ Anchor contact present
- ✅ Chordoma physchem satisfied
- ✅ 4 distinct Bemis–Murcko scaffolds
- ✅ Max Tanimoto to Naar set < 0.6

Slot allocation: 2× A (most validated) · 1× G · 1× F (novel MoA)

Appendix F — Caveats and references

Acknowledged limits

- SPR data has a ~0.65 AUROC ceiling for absolute prediction (batch effects, activity cliffs) → classifier is a ranker, not truth
- Pocket F scores inflated (induced pocket) → pose-only signal
- O0P dual-binder falsifies single-pocket routing; documented fix is multi-assignment
- CheMeleon foundation model tried, dropped for time
- Not chasing sub-µM; target is **credible, fragment-informed, pose-sensible** binders with convergent support

Key numbers

Stage	Output
Downfilter	897,431
API neighbors (unique)	51,560
Combined library	**178,332**
Top per pocket (pre-cluster)	5000
Boltz survivors (site-QC rejects ~90–95%)	<100 per pocket
After Vina + anchor	final per-pocket shortlist
Submission	**4**

References: Newman et al. *Nat Commun* 2025 · SGC TBXT TEP · PDB 6F59 · onepot CORE
In-repo: `docs/HACKATHON_PLAN.md` · `docs/deployment-model.md` · `docs/docking.md` · `docs/brachyury-site-summary.md` · `docs/pocket-mapper-soundness.md` · `docs/sar-diagnostics-rjg/README.md`